

The Price of Assuming You Will Know

Bayes-adaptive market making under genuine model uncertainty

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Joint model-and-regime planning under persistent uncertainty.

1. Question and result

The original practical trader averages separately optimized, known-model action values. It learns from public data during execution, but its planning score assumes that the model will become known after the current decision. How costly is that shortcut when regime persistence is uncertain? This extension implements a joint model-and-regime Bellman recursion, derives model-revelation relaxations and compares executable policies on fresh, paired market trajectories.

Prespecified cold-start result. Joint-belief minus weighted-Q net objective is **+0.004498**, with an empirical paired 95% interval **[-0.001066, +0.010062]** per 30-decision episode. The prespecified economically meaningful improvement criterion is not met. The threshold is 0.002, the fee on one market execution excluding its spread. Sampling uncertainty and numerical approximation error are different: this interval describes the implemented policies, not a certified optimality gap.

The experiment does not establish the registered economically meaningful improvement. The observed estimate and interval are retained even if other comparisons look more favorable.

Fixed setting. The model has $\theta = .35$ and unknown $\kappa \in \{.002, .10\}$ with an equal prior, $T = 30$ and inventory cap two. Everything else—returns, selected-fill observation law, fees, spread, inventory penalty, safe actions and terminal liquidation—is unchanged. The latent state has four possibilities (M, H) , while the public signal has three values. This is a restricted synthetic experiment, separate from the original campaign.

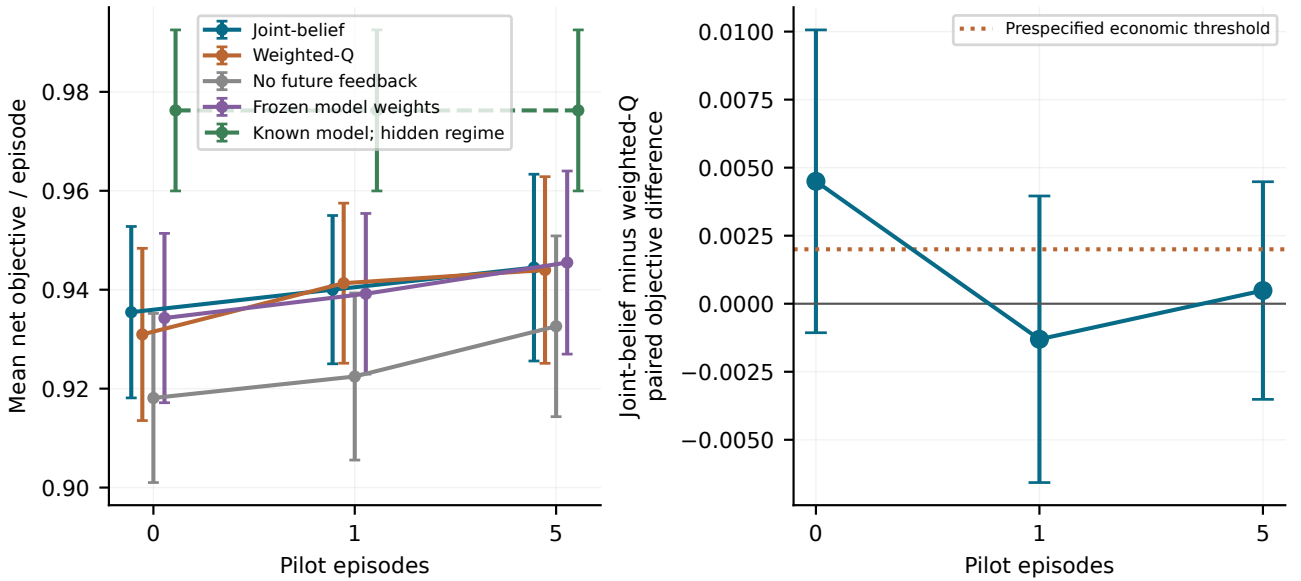


Figure 1. Actual policy performance versus pilot information. Left: policy means with ordinary empirical 95% intervals; the known-model curve is privileged about M only, with H still hidden. Right: paired joint-belief minus weighted-Q differences, using the same trajectories and pilot at each budget. Only the zero-pilot comparison is the primary hypothesis. The two other budgets are prespecified sensitivity conditions. Numerical relaxation values are reported separately on the next page.

What is preserved. The original eight-page main report, eleven-page appendix and canonical outcomes are retained. This extension neither enlarges their deployment population nor revises their reported results. Its conclusions concern only the experiment defined here.

Implementation and computational checks used Codex with Astra and Sol agents.

2. Joint learning and the model-revelation shortcut

Write $p_{j,h} = \Pr(M = j, H_t = h \mid \mathcal{F}_t)$, where M is fixed during an episode and $H_t \in \{-1, +1\}$ switches with probability κ_j . The three coordinates are $w_j = \sum_h p_{j,h}$ and $b_j = \Pr(H_t = +1 \mid M = j, \mathcal{F}_t)$, with $w_2 = 1 - w_1$. The public state is (q, x) ; $\mathcal{A}(q)$ requires every possible fill subset to respect the inventory cap. Only submitted quotes produce fill observations. For observed standardized return $z = (R - .03x)/.30$ and selected outcomes f , the common- θ likelihood is

$$\ell_h(z, f \mid x, a) = \prod_{(s,k) \in a} \Phi\left(\frac{(2f_s - 1)[\tau_{s,k}(x) - h\theta sz]}{\sqrt{1 - \theta^2}}\right), \quad \tau_{s,k}(x) = \Phi^{-1}(p_{s,k}(x)). \quad (1)$$

The empty product is one. Update from H_t 's observation *before* predicting H_{t+1} :

$$\begin{aligned} m_j &= (1 - b_j)\ell_- + b_j\ell_+, \quad m = \sum_j w_j m_j, \quad w'_j = \frac{w_j m_j}{m}, \\ \tilde{b}_j &= \frac{b_j \ell_+}{m_j}, \quad b'_j = \kappa_j + (1 - 2\kappa_j)\tilde{b}_j, \quad p'_{j,+} = w'_j b'_j. \end{aligned} \quad (2)$$

Zero-mass branches have no contribution. Let $r(q, x, p, a)$ denote exact expected marked-to-midpoint wealth change minus the pre-decision $.002q^2$ penalty, including execution fees. For n remaining decisions, terminal liquidation gives $V_0(q, x, p) = -.027|q|$. With $q' = q'(q, a, f)$, the exact recursion is

$$V_n = \max_{a \in \mathcal{A}(q)} \left\{ r(q, x, p, a) + \sum_f \int_{\mathbb{R}} \phi(z) m(z, f) \sum_{x'} K_{xx'} V_{n-1}(q', x', p') dz \right\} \equiv (\mathcal{B}V_{n-1})(q, x, p). \quad (3)$$

The Gaussian density cancels from posterior ratios but remains in this expectation. At cold start $b_1 = b_2 = 1/2$, $m_1 = m_2$ for every action and observation: the first observation has exactly zero model information.

Let $V_{j,n}$ and $Q_{j,n}$ be exact controls knowing $M = j$ while still filtering H . Define U_n^d by revealing M just before decision $d + 1$; for $d \geq n$, there is no useful revelation. Then

$$\begin{aligned} U_n^0(q, x, p) &= \sum_j w_j V_{j,n}(q, x, b_j), \quad U_n^d = \mathcal{B}U_{n-1}^{d-1} \quad (d \geq 1), \\ U_n^1(q, x, p) &= \max_{a \in \mathcal{A}(q)} \sum_j w_j Q_{j,n}(q, x, b_j). \end{aligned} \quad (4)$$

Indeed, conditioning on the common first action and observation gives $mw'_j = w_j m_j$; substituting this into its continuation yields the weighted known-model Q score. Earlier revelation can emulate later revelation by ignoring the extra label. It follows that

$$U_n^0 \geq U_n^1 \geq U_n^2 \geq \dots \geq U_n^n = V_n. \quad (5)$$

These inequalities concern exact planning values. A trader repeatedly maximizing weighted Q receives no actual revelation; its executed value need not equal U_n^1 . Interpolation and quadrature produce approximations to these quantities, not certified numerical bounds. The full proofs and conservative error allowances are in `THEORY.md`.

Approximate planning quantity	Pilot 0	Pilot 1	Pilot 5
U^0 : model known at entry	0.980878	0.973269	0.963642
U^1 : revealed after 1 decision	0.980878	0.973269	0.963642
U^2 : after 2 decisions	0.980616	0.973045	0.963489
U^3 : after 3 decisions	0.980370	0.972820	0.963332
V^{BA} : no revelation	0.938274	0.934593	0.935156

Table 1. Numerical planning values at $q = x = 0$, fresh conditional regime priors $b_1 = b_2 = .5$, and the model posterior fitted from each pilot. Entries average over the balanced pilot population. These are approximate relaxed or Bellman objectives, *not* policy rollout returns and *not* certified numerical upper bounds. U^1 is evaluated directly from weighted known-model Q values; the stored nodal relaxation has a distinct interpolation. The largest observed local hierarchy reversal is 3.1696e-05.

Why the hierarchy is useful. Each delay removes fictitious information while retaining the physical market and admissible actions. The optimizing first action may change. The exact hierarchy applies to prior-weighted values, not individual outcomes or true-model strata. It explains optimism in weighted-Q planning. Uncertified numerical relaxation values and Monte Carlo returns do not certify near-optimality.

At cold start, the numerical U^1 minus achieved weighted-Q mean is +0.049920; delaying revelation to U^3 reduces the corresponding diagnostic to +0.049412. These differences combine planning approximation and Monte Carlo error. They are not certified upper bounds on the heuristic's suboptimality. The complete budget-by-policy diagnostics are exported in the accompanying tables.

3. A fixed experiment that can contradict the hypothesis

Each true model has 30 independent replicates. Every replicate supplies five independent 30-step uniform-safe-action pilot episodes, and 1,000 fresh evaluation episodes. The budgets use no pilot, the first pilot, and all five. The model is fixed within a replicate; every episode starts with a new hidden regime. Evaluation resets the conditional regime beliefs to one half, retains only the pilot model posterior, and never transfers evaluation learning to another episode.

Five policies share each pilot and market tape: joint-belief planning, weighted-Q, no future feedback, frozen future model weights, and the known-model reference. Every feasible policy filters all permitted feedback online. The two suppressions change hypothetical planning updates, not the observations supplied at runtime. Only the privileged reference receives the true model label; it never sees H .

There are **60,000 paired market trajectories** and **900,000 policy-episode records**. Their reuse across policies and pilot budgets does not create additional independent markets. Pairing uses the complete model/replicate/episode/budget key. Within each model, paired episode differences are averaged by replicate; the two stratum means receive equal weight. Welch intervals estimate between-replicate uncertainty. Missing cells, broken pairs, duplicate keys, insufficient replicates and nonfinite outcomes are rejected before inference.

Pilot	Comparison	Objective difference	Net-PnL difference
0	Joint – weighted-Q	+0.00450 [-0.00447, +0.01346]	+0.00771 [-0.00123, +0.01665]
0	Joint – no feedback	+0.01735 [+0.00660, +0.02809]	+0.00969 [-0.00104, +0.02041]
0	Joint – frozen weights	+0.00118 [-0.00558, +0.00795]	+0.00096 [-0.00581, +0.00773]
0	Known model – joint	+0.04079 [+0.02836, +0.05323]	+0.03876 [+0.02635, +0.05116]
1	Joint – weighted-Q	-0.00131 [-0.00979, +0.00718]	+0.00136 [-0.00716, +0.00988]
1	Joint – no feedback	+0.01756 [+0.00666, +0.02846]	+0.01001 [-0.00098, +0.02100]
1	Joint – frozen weights	+0.00083 [-0.00650, +0.00816]	+0.00054 [-0.00678, +0.00785]
1	Known model – joint	+0.03624 [+0.02244, +0.05004]	+0.03480 [+0.02098, +0.04862]
5	Joint – weighted-Q	+0.00048 [-0.00596, +0.00693]	+0.00219 [-0.00428, +0.00866]
5	Joint – no feedback	+0.01186 [-0.00075, +0.02448]	+0.00495 [-0.00773, +0.01763]
5	Joint – frozen weights	-0.00102 [-0.00577, +0.00372]	-0.00125 [-0.00601, +0.00350]
5	Known model – joint	+0.03178 [+0.01420, +0.04936]	+0.03124 [+0.01361, +0.04887]

Table 2. All 24 prespecified supplementary contrasts: four comparisons, three pilot budgets and two metrics. Brackets are Bonferroni-adjusted empirical intervals for this family. The ordinary primary interval is stated on page 1; it is not replaced by a favorable supplementary result. Net PnL includes execution and liquidation costs; net objective additionally deducts the original pre-decision inventory penalty.

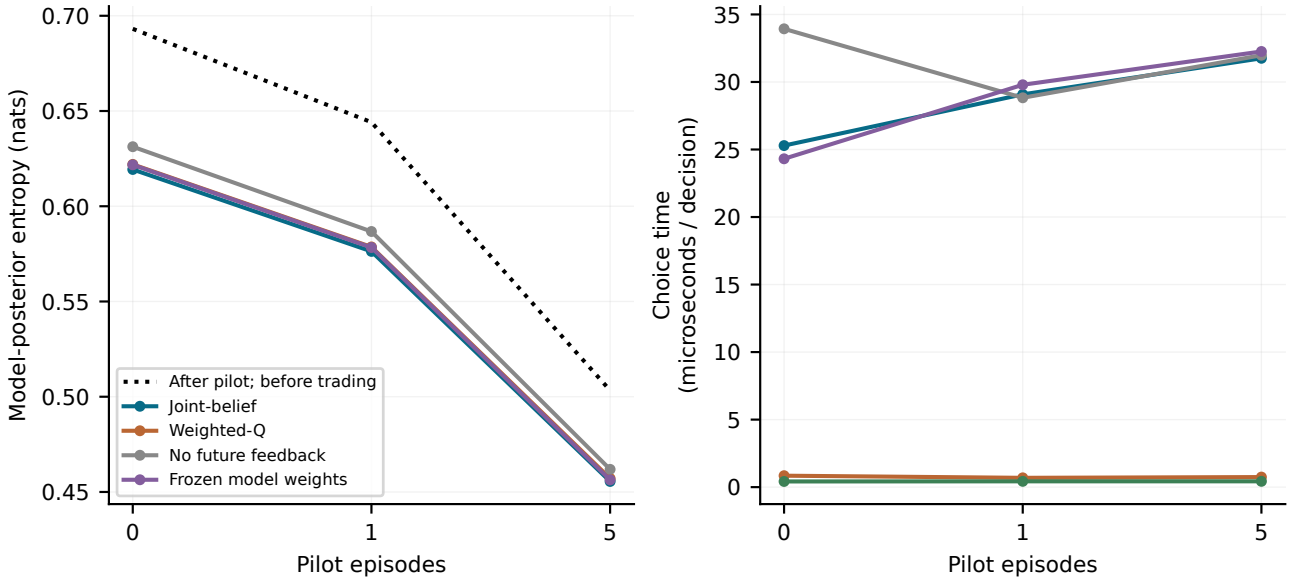


Figure 2. Left: model-posterior entropy after the pilot and after trading; lower entropy alone is not evidence of economic benefit. Right: measured policy-choice time, including table access on this numerical build. These timings are workload observations on a shared machine, not universal latency guarantees. The known-model reference’s shadow public posterior is excluded from the entropy plot because its own model uncertainty is zero.

4. What a controlled decision can establish

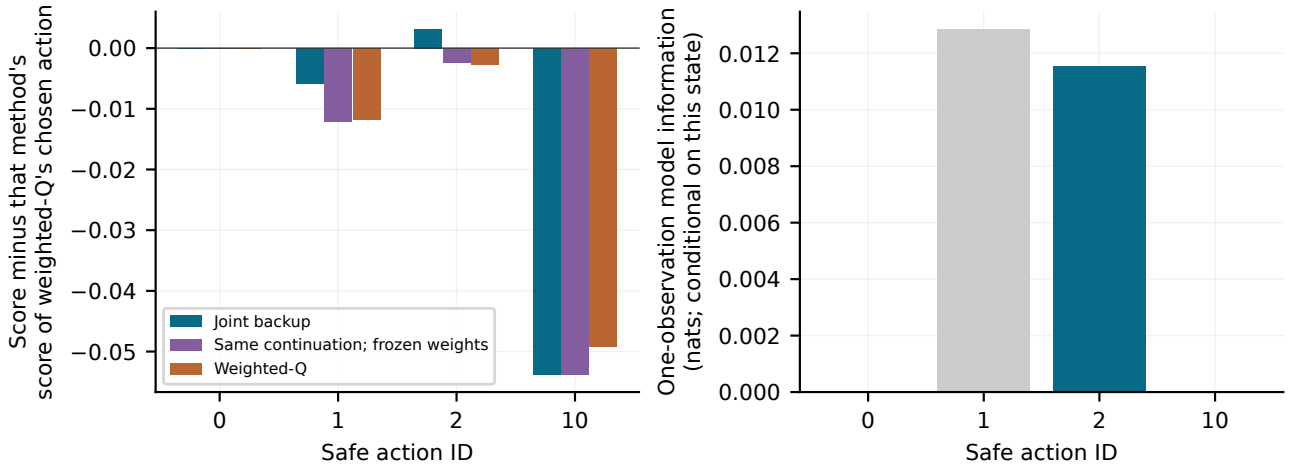
With common θ , an observation depends on the model only through the current hidden regime. Thus $M \rightarrow H_t \rightarrow O_t$ conditional on public history and action, and

$$I(M, H_t; O_t) = I(H_t; O_t) = I(M; O_t) + I(H_t; O_t | M).$$

At cold start, $b_1 = b_2 = .5$: every action's first observation has identical likelihood under both models, so its model information is exactly zero. Regime updating followed by different persistence transitions separates the conditional beliefs; later observations can then identify the model. Parameter information comes from temporal dependence, not a direct model label in a fill.

The explanation search was fixed before evaluation: 60,000 public states reached by uniform-safe actions, plus 77,430 fixed arbitrary belief/horizon probes. It screens action disagreements, ranks by declared score margins and examines a bounded ordered candidate set, without choosing favorable realized PnL paths. Reached and arbitrary-prior states are identified separately.

The fixed search screened 137,430 states and directly checked 512 distinct disagreements. It found 223 cases meeting the complete action-margin and cross-refinement witness criterion. A refinement-stable witness is candidate 48408, with $n = 12$, $q = 2$, $x = 1$ and $p = (0.025081, 0.461728, 0.315619, 0.197572)$. It is reached by the separate public exploration process, not selected from policy evaluation. The direct joint-belief backup selects ask deep (ID 2); weighted-Q selects abstain (ID 0); keeping the same joint-belief continuation but freezing model weights selects abstain (ID 0).



Decision diagnostic. Each score is centered on that same method's score of the weighted-Q action; absolute relaxed and feasible values are not subtracted. Positive information does not itself establish economic benefit.

Quantity	Joint-belief choice	Weighted-Q choice
Immediate expected reward	+0.0460881	+0.0520000
Model information (nats)	+0.0115330	+0.0000000
Regime information conditional on model	+0.0214307	+0.0000000

Attribution limit. The matched diagnostic keeps immediate rewards, outcome probabilities, inventory transitions, conditional regime updates and the same joint-belief continuation fixed. It freezes only the model weights at the next hypothetical planning boundary. This is an artificial computational intervention, not a Bayesian observation deletion or an information bound. When $b_1 \neq b_2$, preserving the likelihood ratio needed by both regime filters also preserves the model Bayes factor:

$$\frac{m_1}{m_2} = \frac{1 - b_1 + b_1 R}{1 - b_2 + b_2 R}, \quad R = \ell_+ / \ell_-.$$

Freezing model weights changes the aggregate regime forecast as well. A stable decision reversal establishes sensitivity to that update; it does not isolate a pure model-information channel. Economic value requires the paired policy comparison, not just entropy reduction or a selected negative immediate reward.

5. Numerical accuracy and reproducibility

The solver uses float64 product coordinates (w, b_1, b_2) , trilinear interpolation in physical probabilities, analytic expected one-step rewards and positive Gauss–Hermite quadrature. Model-weight nodes are uniform; conditional-regime nodes follow $b(u) = \sin^2(\pi u/2)$, concentrating resolution near the endpoints. This numerical-only amendment follows retained uniform-mesh failures and was frozen before local final evaluation, with unchanged probes and tolerances. The selected joint grid is $33 \times 65 \times 65$ with 25 quadrature nodes. The independent known-model scalar references use 321 beliefs and 161 nodes, compared with separate belief and quadrature refinements. The absolute tie tolerance is 10^{-12} ; the lowest safe action ID wins, and the Bellman value uses that selected action’s Q.

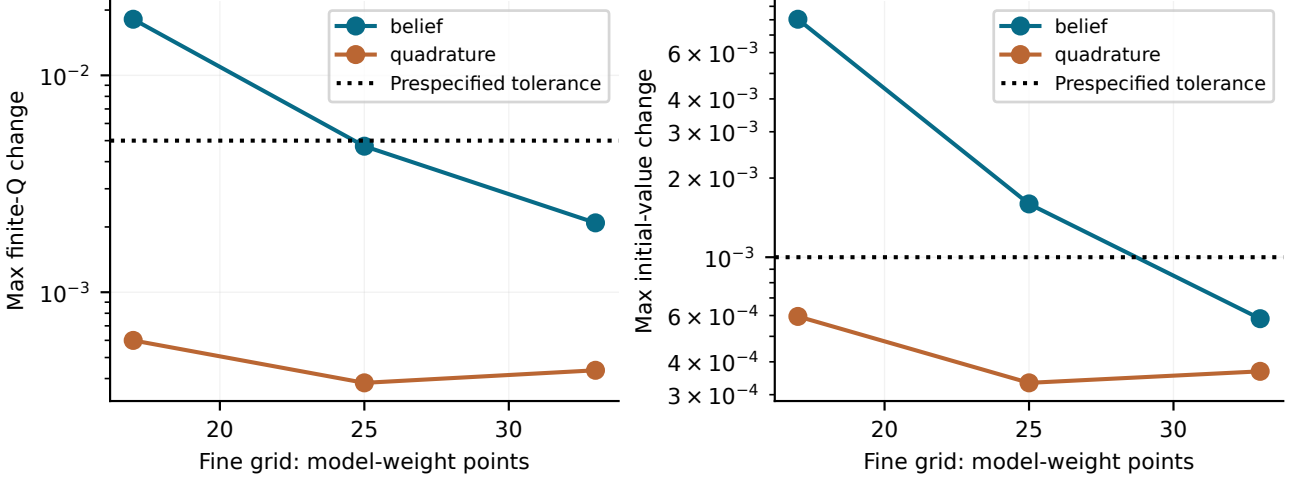


Figure 3. Independent refinement axes, retaining failures. All 30 remaining horizons use identical fixed off-grid probes. Acceptance requires initial value change at most .001, probe value and finite-Q change at most .005, and robust-action disagreement at most .005 when the finer gap exceeds .001. The same rule covers six value modes and all three joint action-score modes.

The final two independently varied axes both satisfy the frozen rule. Earlier failed meshes and the resource-driven V1 interruption remain in the evidence.

For the belief comparison, maximum initial-value, probe-value and finite-Q changes are 0.000583643, 0.00202047 and 0.00208867; the robust-action disagreement fraction is 0.

For the quadrature comparison, maximum initial-value, probe-value and finite-Q changes are 0.000368036, 0.000391736 and 0.000436629; the robust-action disagreement fraction is 0.

The selected-family conservative one-sided allowance at $T = 30$ is 9.46506, and the two-sided off-grid allowance is 15.50378 objective units. They are much larger than .002; their floating evaluation is not an interval-arithmetic certificate.

A proved envelope, not a small optimality certificate. For a fixed policy tree, its expected payoff is linear in the four-state prior. The exact optimal value is therefore convex. The product-grid corner weights have the correct joint-mass barycenter, giving an upper interpolation of exact nodal values. Quadrature error has no known sign, so this fact does not make the computed tables certified upper bounds.

A conservative conditional expected stage-reward bound is $r_* = .231$, with terminal payoff in $[-.054, 0]$. For $D_n = .231n + .027$, exact values satisfy $|V_n(p) - V_n(\bar{p})| \leq D_n \|p - \bar{p}\|_1$. Grid and observation-integration errors can consequently be bounded using the mesh widths and the Wasserstein distance between the normal law and its quadrature measure. The full derivation, computed allowances and omitted floating-arithmetic certification are explicit in [the mathematical appendix](#) and `THEORY.md`. These conservative bounds are too wide to certify the economic threshold. Refinement and independent rollouts give useful numerical evidence, while remaining logically distinct from a uniform mathematical error bound.

6. Validation, interpretation and review entry point

The independent verifier reconstructs all 900,000 outcome records, all 27 million policy decisions, 108,000 posterior-trace rows and 60 pilot sets from the saved primitive tapes and permitted public feedback. It imports neither the producer’s runner nor its statistical routines. All 24 supplementary contrasts, 30 policy means and 180 initial planning-value rows reconcile. The largest absolute ledger discrepancy is 2.50111e-12; the largest public-trace discrepancy is 1.59162e-12.

A fresh process builds the selected family in a previously nonexistent cache. All eleven array-content records and actual file hashes match the reference build. A separate smoke process produces 720 policy records on 48 paired tapes; its pilots, tapes, actions, outcomes, traces, planning values and summaries match exactly, with all 21,600 decisions independently replayed. Measured timing metadata retains its new identity. These are repeated checks of the same seeds, not independent economic observations.

The final economic execution took 10.8 minutes on the recorded stack; the selected-resolution cold reproduction took 28.3 minutes. Both durations are workload observations. The actual review archive has its own packaging and extracted-execution receipt; those checks are separate from numerical and statistical validation. The cold-reproduction duration sums active command execution and excludes the pause for recovering the preserved original archive’s local path. The failed path check and its successful continuation remain in the evidence; no second cold build is claimed.

Interpretation and limits

This extension supplies a different decision recursion and a testable account of the weighted-Q shortcut. It does not claim to invent Bayes-adaptive control: model uncertainty and hidden-state uncertainty are treated jointly in the Bayes-adaptive POMDP literature [1], and information relaxations and approximate POMDP value methods have an established history [2,3]. Our contribution is the specialized model-only revelation interpretation, its implementation in this selected-feedback trading model, and the separately registered evidence.

The experiment does not establish the registered economically meaningful improvement. The observed estimate and interval are retained even if other comparisons look more favorable. The numerical relaxation ladder explains the shortcut’s information optimism, but the present conservative error envelope does not prove a small global optimality gap. The fixed action search supplies 223 qualifying cases; their frequency in a designed search is not an estimate of their prevalence under either policy.

The scope is two correctly specified persistence models with known dependence magnitude, a finite horizon and small inventory. It establishes neither commercial trading validity nor a result under off-grid parameters or arbitrary shifts. Pilot uncertainty is genuine here, but the known finite support remains an assumption. Entropy, relaxed planning optimism and realized policy benefit answer different questions and cannot be substituted for one another.

Reproduction

The public repository contains the protocol, source, numerical refinement, pilots, evaluation tapes, actions, outcomes, posterior traces and statistical tables. `REPRODUCING.md` gives installation and mathematical checks. Four authenticated byte chunks restore the complete compressed ledger.

Recorded evidence belongs to its original numerical build. Fresh runs bind their own source, tables and outputs. Regenerable control arrays are excluded; their specifications and content fingerprints are retained. The small E2E reproduction verifies installation and integration, not the numerical precision of a new full campaign.

References and authorship

1. S. Ross, B. Chaib-draa and J. Pineau (2007), [Bayes-Adaptive POMDPs](#), NeurIPS.
2. M. Hauskrecht (2000), [Value-Function Approximations for Partially Observable Markov Decision Processes](#), JAIR 13, 33–94.
3. R. D. Smallwood and E. J. Sondik (1973), [The Optimal Control of Partially Observable Markov Processes over a Finite Horizon](#), Operations Research 21(5), 1071–1088.